ORIUS: A Professor Review Draft on Universal Runtime Safety for Physical AI under Degraded Observation

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***Abstract*—ORIUS treats degraded observation as the fundamental hidden hazard for Physical AI: controllers act on telemetry, beliefs, and summaries of the world, while safety is judged on the physically true state. This manuscript presents ORIUS, Observation– Reality Integrity for Universal Safety, as a reusable runtime layer that detects reliability collapse, inflates the observation-consistent state set, tightens the admissible action set, repairs or replaces unsafe actions, and emits auditable certificates before release. The paper is organized around the universal runtime claim rather than a witness-row-specific forecasting claim. Its scientific object is a shared runtime grammar grounded in one adapter contract, one benchmark schema, one CertOS governance surface, and one theorem bridge. Battery remains the witness row with the deepest theorem-to-code-to-artifact closure; AV, industrial, and healthcare are defended bounded rows; navigation and aerospace remain explicitly gated by missing real-data closure. The central claim is therefore strong but disciplined: ORIUS establishes a universal runtime safety layer for Physical AI under degraded observation, while equal-domain empirical closure remains a governed target rather than a present-tense fact.**

***Index Terms*—Physical AI safety, degraded observation, observation–action safety gap, runtime assurance, conformal uncertainty, safety filters, cyber-physical systems, certificates**

I. Introduction and Contribution Statement

Physical AI systems do not act on truth. They act on telemetry, inferred state, delayed measurements, missing packets, stale buffers, and learned surrogates of the world. The resulting engineering problem is deeper than forecast error or planner suboptimality: the software stack can remain internally coherent while the physical system becomes externally unsafe. Batteries may violate hidden state-of-charge limits, vehicles may enter unsafe headway regimes on stale perception, industrial plants may cross thermal or pressure envelopes, clinical monitoring pipelines may suppress timely intervention, and navigation or flight systems may remain locally consistent while drifting outside a feasible corridor.

ORIUS is built around the claim that these failures share one universal object: degraded observation changes the semantics of what may still be safely done. The field therefore needs a runtime layer whose job is not to outperform every planner, but to mediate the boundary between degraded observation and physical actuation. ORIUS, Observation–Reality Integrity for Universal Safety, is proposed as that layer. It detects reliability loss, calibrates the observation-consistent state set, constrains the admissible action set, repairs or replaces unsafe candidate actions, and certifies release through an auditable lifecycle. The framework is written around the universal runtime claim rather than a witness-row-specific framing. Battery is retained as the deepest witness row, not as the conceptual center of the claim.

This manuscript positions ORIUS against a precise gap in the literature. Distribution- free inference can quantify uncertainty [1–6]. Runtime assurance and shields can intercept unsafe behavior [7–12]. Safety filters and constrained control can map uncertainty into admissible action sets [13–18]. Fault, anomaly, and drift detection can identify when observation should no longer be trusted [19–24]. What none of these families alone provides is a reusable runtime grammar that links degraded observation to repaired actuation and auditable certificate release across multiple physical domains.

The paper therefore makes four contributions.

1) It defines the observation–action safety gap (OASG) as the right universal hazard object for Physical AI under degraded observation.
2) It presents the Detect–Calibrate–Constrain–Shield–Certify kernel as a typed runtime safety layer with explicit formulas, pseudocode, and certificate semantics.
3) It unifies domain evaluation through one adapter contract, one benchmark schema, and one CertOS governance surface.
4) It argues for a bounded-universal evidence posture: battery is the witness row; AV, industrial, and healthcare are defended bounded rows; navigation and aerospace remain explicitly gated.

A. What ORIUS is and is not

ORIUS is not a new nominal controller, a new world model, or a universal planner. It does not ask every domain to share the same action space, dynamics model, or learning stack. Instead it introduces a narrow but rigorous intervention layer at the moment where candidate actuation is about to be released. That distinction matters. Many physical AI projects already contain estimators, predictors, planners, and local guards. What they usually lack is a reusable semantics for what happens when the controller-

visible state is no longer trustworthy. ORIUS supplies that semantics.

Equally important, ORIUS is not an excuse to hide ad hoc emergency logic behind a new name. Every intervention mode in the framework must be typed and auditable: keep, repair, fallback, degrade, or reject. The release decision is therefore promoted from implementation detail to scientific object. That is the conceptual step that lets the paper talk about batteries, vehicles, industrial plants, healthcare monitoring, and aerospace within one formal vocabulary.

B. Design principles

The framework is organized around four design principles.

1) **Post-nominality:** upstream intelligence remains replaceable; ORIUS is evaluated at the release boundary rather than by forecasting or planning quality.
2) **Typed degradation:** reliability loss, widened uncertainty, repaired action, fallback mode, and certificate status are explicit runtime objects.
3) **Domain-bounded portability:** each adapter owns telemetry semantics and safety geometry, but the kernel and benchmark semantics do not change from one row to the next.
4) **Auditable release:** safety is judged on what actuation is actually released, under what certificate, and with what lifecycle trace.

These principles explain why ORIUS should be read as infrastructure for Physical AI safety rather than as one more domain benchmark result. A battery controller can be good or bad; an AV planner can be good or bad; an industrial predictor can be good or bad. ORIUS asks a different question: when those nominal systems become observation- degraded, what runtime object decides whether the next action remains legally releasable?

The draft is intentionally ambitious but disciplined. It uses a universal writing posture, but it keeps one explicit evidence-boundary section in the main body. It includes formulas, theorem statements, proof sketches, runtime pseudocode, decisive results tables, and cross-domain synthesis, while pushing proof-depth and artifact-depth material into two separate appendices.

II. Problem Definition: Degraded Observation and OASG

Consider a discrete-time physical system observed through a degraded telemetry channel:

$$x_{t+1} = f(x_t, a_t, \omega_t), \qquad y_t = h(x_t, \nu_t), \qquad \hat{x}_t = \eta(y_{0:t}), \tag{1}$$

where $x_t \in \mathcal{X}$ is the true state, y_t is the observation packet, $\hat{x}_t$ is the controller-visible state, and $a_t^{\text{cand}} = \pi_t(\hat{x}_t)$ is the candidate action produced by the nominal controller. Let $\mathcal{S} = \{x \in \mathcal{X} : \phi(x) \geq 0\}$ denote the safe set, and let $\mathcal{A}(x)$ denote the set of actions that preserve safety when the true state is x.

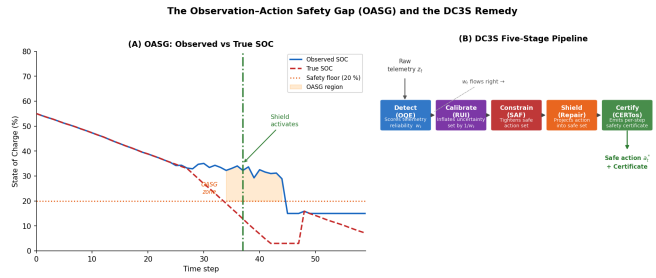

Fig. 1: Degraded observation turns controller legality into a latent safety hazard: the candidate action can be valid on the observed state while unsafe on the true state.

The observation–action safety gap is the event

$$\text{OASG}_t \equiv \mathbf{1}\big[a_t^{\text{cand}} \in \mathcal{A}(\hat{x}_t) \ \wedge \ a_t^{\text{cand}} \notin \mathcal{A}(x_t)\big]. \tag{2}$$

The key point is that OASG is an action-semantics failure, not only a state-estimation failure. A better estimator can reduce its frequency, but the safety event itself is defined by the legality of released actuation under degraded observation.

ORIUS turns this hazard into a runtime object by making observation quality explicit. Let $w_t \in [0, 1]$ be a reliability score computed from freshness, missingness, sensor disagreement, residual behavior, or domain-specific degradation signatures. The observation-consistent set is then inflated as

$$U_t = \{x \in \mathcal{X} : d_t(x, \hat{x}_t) \leq \beta_0 + \lambda(1 - w_t) + \delta_t\}, \tag{3}$$

where β_0 is the nominal uncertainty radius, $\lambda(1 - w_t)$ is the reliability-conditioned inflation, and δ_t captures drift- or history-based correction. The universal contract does not insist on a single metric d_t; it insists only that the set widens monotonically when reliability falls.

The corresponding tightened safe-action set is

$$\mathcal{A}_t^{\text{safe}}(U_t) = \{a \in \mathcal{A} : g(x, a) \leq 0, \ \forall x \in U_t\}, \tag{4}$$

where $g(x, a)$ is the domain safety constraint. The released action is then

$$a_t^{\text{safe}} = \begin{cases} a_t^{\text{cand}}, & a_t^{\text{cand}} \in \mathcal{A}_t^{\text{safe}}(U_t), \\ \mathcal{R}_t(a_t^{\text{cand}}, \mathcal{A}_t^{\text{safe}}(U_t)), & \mathcal{A}_t^{\text{safe}}(U_t) \neq \varnothing, \\ \mathcal{F}_t(c_t), & \mathcal{A}_t^{\text{safe}}(U_t) = \varnothing, \end{cases} \tag{5}$$

with repair map $\mathcal{R}_t$, fallback policy $\mathcal{F}_t$, and certificate context c_t.

This yields the central runtime question for the paper: once degraded observation is acknowledged, what set of actions remains certifiably safe and auditable? ORIUS answers that question with one runtime kernel, one benchmark schema, and one governance discipline. The theorem bridge later shows that the resulting safety layer is not a heuristic wrapper but a structured bound-preserving construction under explicit assumptions.

The same object also governs evaluation. ORIUS measures safety on the true state rather than on the observed

TABLE I: Canonical runtime objects for OASG-aware release.

Object	Role
w_t	Reliability score derived from freshness, missingness, and disagreement.
U_t	Observation-consistent state set widened when trust falls.
$\mathcal{A}_t^{\text{safe}}(U_t)$	Tightened action set that remains legal for every state in U_t.
$\mathcal{R}_t$	Repair map used when the candidate action is unsafe but the safe set is non-empty.
$\mathcal{F}_t$	Explicit fallback used when no repaired safe action exists.
c_t	Certificate payload recording margin, reliability, release mode, and audit continuity.

state alone. At the episode level this yields the true-state violation rate

$$\text{TSVR} = \frac{1}{T}\sum_{t=0}^{T-1} \mathbf{1}[x_t \notin \mathcal{S}], \tag{6}$$

and the runtime intervention rate

$$\text{IR} = \frac{1}{T}\sum_{t=0}^{T-1} \mathbf{1}[a_t^{\text{safe}} \neq a_t^{\text{cand}}]. \tag{7}$$

These metrics make the claim falsifiable. A system cannot claim degraded-observation safety if it only looks legal on the observed state, or if it relies on undocumented emergency overrides rather than explicit runtime interventions.

III. Related Work and the Insufficiency of Prior Families

ORIUS does not claim that prior work ignored uncertainty or safety. The claim is narrower and stronger: the major adjacent families solve pieces of the problem, but not the full runtime bridge from degraded observation to repaired actuation and auditable release.

A. Conformal prediction and calibration are necessary, but not enough

Distribution-free and conformal methods provide finite-sample uncertainty surfaces for learned predictors [1–6, 25]. These tools are indispensable for ORIUS because the runtime must widen the observation-consistent set when reliability drops. But they do not, by themselves, specify which action remains legal after that widening. A prediction interval is not a certificate of actuation legality. ORIUS therefore treats calibration as the input to a safety layer rather than as the safety layer itself.

B. Runtime assurance and Simplex-style supervision do not close the observation gap

Runtime assurance, supervisory control, and Simplex-style architectures [7–12, 26] show that nominal intelligence should not be the final authority over actuation. ORIUS inherits that insight directly. The gap is that many supervisory designs begin after a candidate action already exists and assume that the state estimate is coherent enough to be monitored. ORIUS starts earlier: it treats the observation surface itself as unsafe, which forces the runtime to reason jointly about reliability, uncertainty inflation, repair geometry, fallback, and certificate validity.

C. Safety filters and constrained control are mathematically strong but plant-local

Barrier functions, robust MPC, tube MPC, predictive safety filters, and reachability- based controllers can turn model uncertainty into admissible action sets [13–18, 27, 28]. ORIUS does not compete with that mathematics. It packages those ideas behind a typed runtime contract that can host multiple geometries across multiple plants. The scientific move is not a new universal controller; it is one reusable way to make degraded observation alter actuation semantics.

D. Anomaly, drift, and fault detection stop too early

Change detection, drift detection, anomaly scoring, and CPS fault analysis provide the language for recognizing when telemetry should no longer be trusted [19–24, 29, 30]. Yet many systems stop at an alarm or a reliability score. ORIUS treats that as an incomplete runtime response. Once trust has fallen, the system still needs to decide which actions remain legal, which actions must be repaired, when fallback is required, and how that decision is certified.

E. Domain-specific safety work rarely becomes a universal runtime grammar

Energy dispatch, autonomous driving, industrial control, clinical monitoring, navigation, and aerospace safety each contain strong plant-level ideas, but most remain tied to one domain's state geometry, one benchmark, and one intervention semantics. The result is local excellence without a shared runtime object. ORIUS positions itself as a cross-domain safety-layer grammar that can sit above those domain-specific methods while keeping the data and proof posture explicitly tiered.

The ORIUS thesis is therefore integrative but not vague. The universal object is not "safe AI" in the abstract. It is the typed runtime layer that binds reliability, uncertainty, admissible actions, repaired release, and certificate governance under one action-semantics problem.

IV. ORIUS Runtime Architecture

ORIUS is a post-nominal runtime layer. It does not replace the upstream predictor, planner, optimizer, or policy. It wraps them with a narrow contract that begins at raw telemetry and ends at certificate-gated release. This is the architectural reason the framework can remain domain-agnostic even when the domains do not share nominal models or action geometry.

The kernel consists of five stages.

TABLE II: Why prior families remain necessary but insufficient for the ORIUS object.

Method family	What it contributes	Why it is insufficient for OASG
Conformal and adaptive calibration	Distribution-free uncertainty and shift-aware coverage control	Does not decide repaired actuation, fallback legality, or certificate release after trust degrades.
Runtime assurance and shielding	Supervisory interception and safety-controller fallback	Often assumes the observation surface is coherent enough to supervise directly.
Safety filters and robust constrained control	Strong admissible-set mathematics under uncertainty	Usually remains plant-local and does not define a reusable benchmark/governance contract.
Anomaly, drift, and fault detection	Detects that telemetry has become unreliable	Stops at alarms or scores unless paired with repair, fallback, and audit semantics.
Domain-specific CPS safety stacks	High-quality local solutions in one plant family	Rarely expose a universal runtime grammar that transfers across domains without changing the scientific object.

1) **Detect:** estimate reliability w_t and drift flags from freshness, missingness, cross-sensor consistency, residual behavior, and domain-specific diagnostics.
2) **Calibrate:** inflate the observation-consistent set U_t rather than trusting a nominal point estimate.
3) **Constrain:** recompute the safe action set under the widened state set.
4) **Shield:** keep, repair, or replace the candidate action.
5) **Certify:** emit a causal certificate and update the governance state.

A. Execution path through the kernel

The practical execution path is simple enough to be implemented in a few typed calls, but rich enough to support formal analysis. A telemetry packet enters through the domain adapter with time stamps, missingness indicators, residual channels, and any domain-specific diagnostics. The detect stage converts those observations into a reliability score and degradation flags. The calibrate stage then refuses the usual engineering shortcut of keeping a point state estimate when trust has fallen; instead it materializes the observation-consistent state set that the controller must now respect. The constrain stage computes what actuation remains safe for every state in that set. Shielding then either preserves the candidate action, repairs it into the safe set, or invokes an explicit fallback. Certify binds the release decision to a certificate payload and governance state transition.

That same execution path has different physical interpretations across domains. In battery systems it becomes charge or discharge limitation under hidden state-of-charge uncertainty. In AV it becomes governed braking or maneuver restriction under degraded headway confidence. In healthcare it becomes intervention legality under uncertain vital-sign trust. The point of ORIUS is not that these domains share dynamics. The point is that they share a release-boundary problem once degraded observation is made explicit.

The kernel becomes scientific rather than architectural once its objects are written explicitly. ORIUS uses a reliability-conditioned uncertainty envelope

$$U_t = \{x : m_t(x, \hat{x}_t) \le q_{1-\alpha_t}(w_t, \Delta_t)\}\,, \tag{8}$$

Algorithm 1 Detect–Calibrate–Constrain–Shield–Certify Runtime Step

Require: observation packet o_t, history $\mathcal{H}_t$, candidate action a_t^{cand}
compute reliability score w_t and degradation flags from o_t
build observation-consistent set U_t using reliability-aware inflation
construct tightened safe-action set $\mathcal{A}_t^{\text{safe}}(U_t)$
if $a_t^{\text{cand}} \in \mathcal{A}_t^{\text{safe}}(U_t)$ **then**
  $a_t^{\text{safe}} \leftarrow a_t^{\text{cand}}$
else if $\mathcal{A}_t^{\text{safe}}(U_t) \neq \varnothing$ **then**
  $a_t^{\text{safe}} \leftarrow \mathcal{R}_t(a_t^{\text{cand}}, \mathcal{A}_t^{\text{safe}}(U_t))$
else
  $a_t^{\text{safe}} \leftarrow \mathcal{F}_t(c_t)$
end if
emit certificate c_t with $(w_t, U_t, \gamma_t, \texttt{mode}, \texttt{audit_hash})$
release a_t^{safe} only if certificate is valid or fallback is explicit

where m_t is a domain-specific mismatch measure and $q_{1-\alpha_t}$ is a reliability-aware quantile or inflation operator. The tightened action set is

$$\mathcal{A}_t^{\text{safe}}(U_t) = \{a \in \mathcal{A} : g(x, a) \le 0,\ \forall x \in U_t\}, \tag{9}$$

and the certificate-relevant margin is

$$\gamma_t(a) = \inf_{x \in U_t} \phi\big(f(x, a, 0)\big), \tag{10}$$

so that $\gamma_t(a) \ge 0$ indicates one-step safety for every state in the observation-consistent set.

The three equations above are the core of the framework. The first says that trust in observation is itself a runtime variable. The second says that admissible actuation is defined against a set, not only against a point estimate. The third says that the certificate is not a paperwork artifact; it is tied to a concrete safety margin over that set. Together they turn degraded observation from a monitoring nuisance into a first-class control-theoretic and systems object.

This algorithm is the real reusable object in ORIUS. It explains why the framework can support batteries, vehicles, industrial plants, healthcare monitoring, navigation, and aerospace without pretending those domains share the same

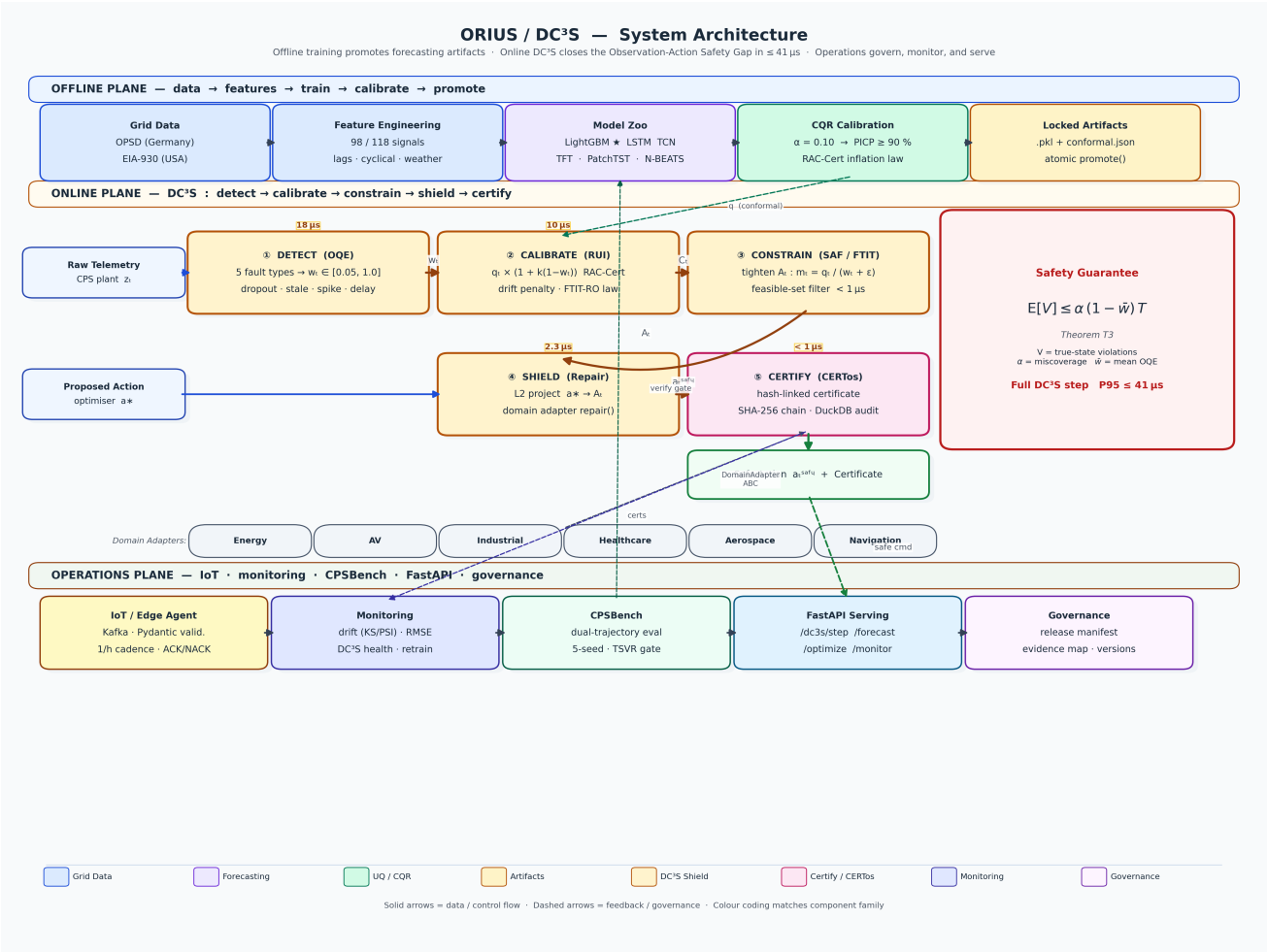

Fig. 2: ORIUS as a universal runtime layer between nominal intelligence and physical actuation. Domain-specific adapters feed a shared Detect–Calibrate–Constrain–Shield– Certify kernel whose outputs are benchmarked and governed under one contract.

nominal control law. What is shared is the runtime grammar that mediates degraded observation before actuation.

Two design choices keep this object scientifically clean. First, ORIUS is intentionally narrow: it sits between nominal intelligence and release rather than trying to subsume forecasting, planning, or identification. Second, the runtime layer is typed all the way through. Reliability, uncertainty, repair mode, fallback mode, certificate state, and audit continuity are explicit objects. That is why the framework can be benchmarked and governed as a reusable safety layer instead of described only as a software pattern.

B. *Why the architecture is universal*

ORIUS achieves universality by freezing the meaning of a few objects and refusing to freeze everything else. The meaning of reliability, uncertainty widening, tightened action legality, repaired release, fallback, and certificate validity is invariant across domains. Telemetry parsers, nominal controllers, local geometry, and domain constraints are not invariant and are deliberately delegated to adapters. This split is what lets the same runtime layer sit above an optimizer in energy systems, a planner in autonomy, a control stack in process industries, or a monitored intervention pipeline in healthcare without becoming mathematically empty.

V. Universal Contracts: Adapter, Benchmark, and CertOS Governance

ORIUS is organized around three contracts: the adapter contract, the benchmark contract, and the CertOS governance contract. Together they explain how degraded observation is converted into a typed runtime safety layer rather than a battery-only control recipe.

A. *Adapter contract*

The adapter is the only place where domain-specific telemetry and constraints are allowed to enter the universal runtime. The repository and its supporting design documents make this boundary explicit: `src/orius/certos/runtime.py` provides a domain-neutral governance surface, `docs/UNIVERSAL_BENCHMARK_SPEC.md`

defines the common step schema, and `docs/UNIVERSAL_GOVERNANCE_SPEC.md` requires a domain governance policy instead of battery-shaped runtime state.

In practical terms, each domain adapter must expose a true-state violation predicate, an observed-state satisfiability predicate, an optional margin computation, a fallback action, and certificate-compatible action repair. That is the minimum needed to make ORIUS portable across battery energy storage, autonomous vehicles, industrial control, healthcare monitoring, navigation, and aerospace without changing the meaning of safety at the kernel level.

The adapter can therefore be read as a typed semantic projection

$$\mathfrak{D}_t : (o_t, \mathcal{H}_t, a_t^{\text{cand}}) \mapsto (w_t, U_t, \mathcal{A}_t^{\text{safe}}, \mathcal{R}_t, \mathcal{F}_t, c_t),$$

where telemetry and nominal actions are translated into the universal runtime objects that ORIUS knows how to govern. This is one of the most important clarifications in the paper: universality does not mean a universal plant model. It means one typed safety projection from domain observation to governed actuation.

B. Benchmark contract

ORIUS-Bench is the manuscript-facing replay contract. The universal step record keeps the same fields across domains: `true_constraint_violated`, `observed_constraint_satisfied`, `true_margin`, `observed_margin`, `intervened`, `fallback_used`, `certificate_valid`, `latency_us`, and `domain_metrics`. The metrics family is equally fixed: TSVR, OASG, certificate validity accuracy, intervention rate, audit completeness, latency percentiles, and coverage diagnostics when conformal uncertainty is active.

The benchmark contract matters because it decides what evidence counts. A step is not judged by whether the nominal controller felt confident. It is judged by whether the true constraint was violated, whether the observed state looked safe, whether ORIUS intervened, whether fallback was used, whether the certificate remained valid, and what latency budget was consumed. This prevents the evaluation from collapsing into prediction accuracy or task reward while still remaining agnostic to the nominal model.

This contract prevents the argument from collapsing into a single battery result table. It also keeps the evidence boundary honest: the same schema can be reused across all six rows, but the rows do not carry the same proof depth.

C. CertOS governance contract

CertOS is the runtime governance layer. Its role is deliberately narrow: do not release an action unless the runtime can either validate a current certificate or replace the action with an explicit fallback. The policy surface is domain-neutral and consists of five hooks: `is_actuation(action)`, `fallback_action(constraints, state)`, `horizon_update(certificate, reliability, drift, constraints, validity_horizon)`, `required_certificate_fields()`, and `integrity_projection(action)`.

Algorithm 2 CertOS Certificate Release and Lifecycle Update

Require: repaired action a_t^{safe}, certificate candidate c_t, reliability w_t, constraints κ_t
  project a_t^{safe} through `integrity_projection`
  if `required_certificate_fields()` missing **then**
    mark certificate invalid and request fallback
  end if
  validate lifecycle state and audit hash continuity
  if certificate valid **then**
    compute horizon update using `horizon_update`
    append validated certificate to audit ledger
    release projected action
  else
    replace with `fallback_action` and append degraded certificate
    release fallback only if lifecycle invariants still hold
  end if

TABLE III: The three public ORIUS contracts used throughout the professor draft.

Layer	Contract obligation
Adapter	Convert degraded telemetry into reliability, uncertainty, safe-action, repair, fallback, and certificate objects.
Benchmark	Keep one true-state safety schema and one metric family across all domains.
CertOS	Release only validated or explicit fallback actions with audit continuity.

The invariants are correspondingly small and auditable: no actuation without a valid or fallback certificate path, an unbroken certificate hash chain, and fallback only when validity collapses. That is the right level of abstraction for a professor-facing draft that wants to argue for a fundamental safety layer.

This is also where ORIUS separates itself from many safety wrappers that are difficult to compare scientifically. If a runtime layer cannot explain what is released, under what certificate, for how long that certificate remains valid, and how fallback is recorded, then it is hard to distinguish principled safety governance from an implementation-specific patch. CertOS is deliberately narrow so that those questions are answerable.

The claim boundary is therefore straightforward: ORIUS is a universal runtime safety contract for Physical AI under degraded observation, but the evidence remains tiered by domain and is not presented as equal-depth proof across all rows.

That methodological discipline matters as much as the API itself. It prevents the paper from reading like a battery controller with extra documentation. By freezing the adapter boundary, the benchmark record, and the governance hooks, ORIUS defines one reproducible scientific

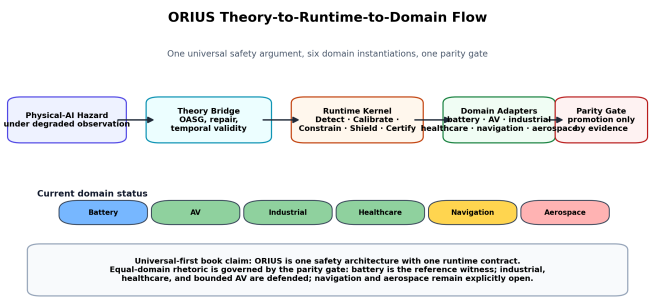

Fig. 3: The theory bridge separates three levels of the claim: formal runtime objects, domain-specific evidence obligations, and promotion status under the parity gate.

object whose meaning does not change from one domain chapter to the next.

VI. Theory Bridge and Universality Gate

This section distills the formal spine of ORIUS for a professor-facing IEEE draft. It connects the battery proof ladder, the universal kernel, and the cross-domain promotion logic without reintroducing the full monograph narrative. The key point is that ORIUS is not presented as a battery theorem generalized by analogy. It is presented as a runtime safety grammar whose transfer obligations are explicit, checkable, and separate from the evidence hierarchy.

A. Formal setting

We consider a partially observed safety-constrained control system on a discrete horizon $t = 0, 1, \ldots, T-1$:

$$x_{t+1} = f(x_t, a_t, \omega_t), \qquad y_t = h(x_t, \nu_t), \qquad w_t = \psi(y_{0:t}),$$

with true state $x_t \in \mathcal{X}$, action $a_t \in \mathcal{A}$, observation history $\mathcal{H}_t = \sigma(y_{0:t})$, and reliability score $w_t \in [0, 1]$. The safe set is

$$\mathcal{S} = \{x \in \mathcal{X} : \phi(x) \geq 0\},$$

and the episode violation count is

$$V_T = \sum_{t=0}^{T-1} \mathbf{1}[x_t \notin \mathcal{S}].$$

The ORIUS runtime layer takes a nominal candidate action $a_t^{\text{cand}} = \pi_t(\mathcal{H}_t)$, constructs an observation-consistent state set U_t, tightens the admissible action set $\mathcal{A}_t^{\text{safe}}(U_t)$, repairs or replaces the candidate action, and emits a causal certificate.

B. Universality gate

The equal-domain ambition of ORIUS is governed by a universality gate. A domain instantiation is not promoted by rhetorical similarity; it is promoted when it supplies:

1) a declared observation model and a reproducible replay or experimental protocol;
2) a sound observation-consistent set U_t and a domain-specific safe-action set;
3) a repair or fallback operator that is measurable, causal, and certificate-backed;
4) a benchmark compatible with the universal schema and a certificate that can be audited;
5) a domain-specific safety reduction result that can be checked against the shared contract.

This gate is the manuscript's main discipline rule: architecture may be universal before evidence is equal, but the equal-domain claim is only allowed once the gate is passed.

Proposition VI.1 (Universality gate)**.** *If a domain instantiation satisfies the universal adapter contract and discharges the promotion obligations above, then the ORIUS kernel can be interpreted on that domain as a runtime safety layer with the same formal objects used in the battery reference proof: observation reliability, uncertainty inflation, admissible repair, fallback, and certificate semantics. If those obligations are not discharged, the manuscript may retain only a bounded or gated claim.*

Proof sketch. The proposition is a packaging statement, not a new control theorem. It follows directly from the typed kernel and adapter surface described in the universal-completeness chapter: the runtime only requires an observation packet, a reliability score, an observation-consistent state set, a tightened safe-action set, a repair operator, a fallback, and a causal certificate. Appendix A lists these objects and their assumption dependencies. Once a domain supplies the same objects, the formal language of the battery ladder applies without changing the runtime contract; what changes is the evidence status of the domain. □

C. T9: Reliability-aware repair is structurally necessary

The first theoretical claim is that ignoring reliability is not a benign simplification. Under persistent observation degradation, a controller that does not adapt its safety margin to w_t cannot eliminate violations by algorithmic tuning alone.

Theorem VI.2 (T9: Reliability-aware repair is necessary)**.** *Assume persistent observation degradation with mean rate $d > 0$, a φ-mixing fault process, and the standing boundary-reachability assumptions used in the battery reference proofs. Then there exists a constant $c > 0$ such that any controller that does not adapt its safety margin to w_t satisfies*

$$\mathbb{E}[V_T] \geq c\, d\, T.$$

Proof sketch. The battery proof surface shows the core mechanism: if the controller keeps a fixed safety margin while the telemetry channel degrades, then the observed state can remain legitimate even as the true state crosses the constraint boundary. The full argument uses separated windows and a martingale concentration step to convert per-window violation probability into a linear-in-T lower bound. The essential point is that the degradation process,

not the nominal controller class, determines the scaling law. □

D. T10: Degraded observation induces a reliability-risk frontier

The second claim is a lower-bound statement: degraded observation creates a nontrivial risk frontier that cannot be removed by a stronger controller alone.

Theorem VI.3 (T10: Reliability-risk frontier)**.** *Suppose that for each step t there is a boundary-testing subproblem with safe and unsafe hypotheses $H_{0,t}$ and $H_{1,t}$, and that the induced observation laws satisfy $\mathrm{TV}(P_{0,t}, P_{1,t}) \leq w_t$. If the unsafe-side prior satisfies $\Pr(H_{1,t}) \geq p_t$, then every controller π obeys*

$$\mathbb{E}[V_T(\pi)] \geq \frac{1}{2} \sum_{t=1}^{T} p_t(1 - w_t).$$

In particular, if $p_t \geq \alpha/2$ for all t, then

$$\mathbb{E}[V_T(\pi)] \geq \tfrac{1}{4}\alpha(1 - \bar{w})T.$$

Proof sketch. This is a Le Cam style reduction. Each step becomes a binary decision problem between a safe and unsafe boundary neighborhood. The smaller the total variation gap between the two observation laws, the less information the controller has about which side of the boundary it is on. The lower bound then follows by multiplying the irreducible classification error by the mass of the unsafe hypothesis and summing over the horizon. The result is intentionally stylized: it isolates the scaling law, not a globally sharp minimax constant. □

E. T11: Structural transfer through the universal adapter contract

The third claim is the one that matters most for a universal paper draft. It says that the battery proof pattern can transfer if, and only if, the domain exposes the right runtime objects.

Theorem VI.4 (T11: Structural transfer)**.** *Let $\mathcal{D}$ be any domain instantiation that satisfies the universal adapter contract: it provides an observation packet, a reliability score w_t, an observation-consistent state set U_t, a domain-safe action set $\mathcal{A}_t^{\mathrm{safe}}(U_t)$, a repair map, and a fallback action. If $\Pr[x_t \in U_t \mid \mathcal{H}_t] \geq 1 - \alpha$ and every repaired action lies in the tightened safe-action set, then*

$$\Pr[x_{t+1} \in \mathcal{S} \mid \mathcal{H}_t] \geq 1 - \alpha.$$

Proof sketch. Condition on the observation history. On the event $x_t \in U_t$, soundness of the tightened safe-action set implies that every admissible repaired action preserves the safe set at the next step. If the tightened set is empty, the fallback action preserves safety by contract. Therefore the only remaining source of failure is the probability that the true state lies outside U_t, which is at most α. Appendix A and the theorem-surface register record the battery proof instances that instantiate this logic in the reference domain. □

Corollary VI.5 (Episode aggregation under a domain risk budget)**.** *If a domain supplies a predictable per-step risk budget r_t such that $\Pr[x_{t+1} \notin \mathcal{S} \mid \mathcal{H}_t] \leq r_t$, then*

$$\mathbb{E}[V_T] \leq \sum_{t=1}^{T} r_t.$$

When the budget has the battery-style envelope $r_t \leq \alpha(1 - w_t)$, the familiar upper bound $\mathbb{E}[V_T] \leq \alpha(1 - \bar{w})T$ follows immediately.

F. Interpretation for the IEEE draft

The practical message for the IEEE manuscript is that the theory bridge does not need to be a second thesis chapter. It needs to do three things well: state the universal objects, state the universality gate, and state the proof skeleton that connects the reference battery ladder to any new domain that clears the adapter contract. That is the strongest possible version of the claim without inventing unsupported empirical parity.

This is why the theory bridge belongs in the main paper instead of disappearing into a supplement. The ORIUS claim is not merely that a benchmark improved. It is that degraded observation changes the structure of what safe release means, and that the runtime layer makes that change explicit in a way that can be proved, benchmarked, and audited.

VII. Battery Witness Domain

The battery row is the strongest witness in ORIUS because it closes the entire chain from assumptions and theorems to controller code, replay artifacts, latency budgets, and certificate traces. It is therefore the right place to show the runtime grammar in its deepest form without allowing the paper to collapse back into a battery-only manuscript.

A. Problem, state, action, and safety object

The witness task is storage dispatch under uncertain load, renewable injection, price signals, and telemetry faults. The true state includes at least state of charge, operational bounds, and dispatch context; the candidate action is charge/discharge power. The governing safety object is not cost. It is hidden state violation under degraded observation, measured on the true state rather than on controller-visible legality.

B. Data and replay surface

The battery evidence is built on the locked CPSBench-derived replay surface with 168-hour episodes, fixed seeds, and fault families that include dropout, stale sensing, delay, drift, and mixed degradation. This is what makes the battery row useful as a witness: it is a repeatable degraded-observation environment rather than a one-off case study.

TABLE IV: Battery witness summary from the tracked main battery result surface.

Controller	TSVR	IR (%)	Cost Δ (%)
ORIUS FTIT	0.0	8.8	14.8
ORIUS wrapped	0.0	0.0	5.5
Adaptive conformal + repair	0.0	2.1	2.3
Deterministic LP	0.1	0.0	0.0
Robust fixed interval	0.2	0.0	-0.9

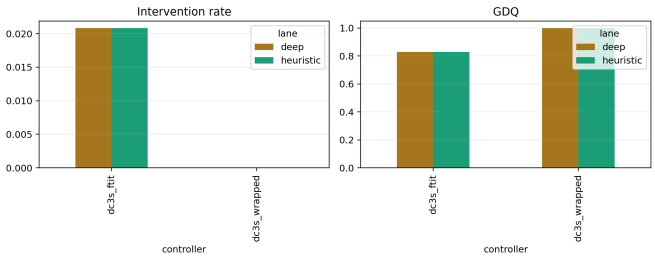

Fig. 4: Battery witness comparison for reliability-conditioned ORIUS variants. The important signal is structural: repair-aware variants preserve zero true-state violations under degraded telemetry.

C. Witness result

The decisive battery result is that ORIUS repair-aware controllers eliminate true-state violations under the locked protocol while non-repair baselines do not. The deep OQE variant is not presented as a miracle predictor; it is a quality signal routed into a repair-and-certify layer that prevents hidden violations.

The deeper trace, ablation, and raw-sequence evidence remains important, but it belongs in Appendix B. In the main paper the battery row serves one purpose: it proves that the runtime safety layer is not merely architectural. It changes the released-action semantics and removes true-state failures on the strongest witness surface currently available in the repo.

VIII. Runtime and Empirical Results

The decisive empirical question for ORIUS is not whether one domain achieves the lowest cost or the smallest interval width. It is whether the runtime layer materially reduces true-state violations under degraded observation while remaining operationally cheap and auditable. The current repository truth supports that claim most strongly in the battery witness row and then at bounded depth across AV, industrial, and healthcare.

The battery witness establishes the strongest result surface: ORIUS repair-aware controllers eliminate true-state violations under the locked degraded-telemetry protocol while maintaining bounded intervention rates. The cross-domain runtime evidence then shows that the same runtime grammar remains cheap enough to be plausible across non-energy domains. The important result is therefore joint: safety repair is doing real work, and the added runtime layer is not computationally extravagant.

Three results matter most.

1) **Safety reduction:** the battery witness row is the clearest proof that reliability-conditioned repair prevents hidden violations that remain visible under deterministic and static-interval baselines.
2) **Runtime plausibility:** the defended rows remain in the sub-millisecond regime at the P95 level, which means ORIUS behaves like a lightweight safety layer rather than a second controller.
3) **Governed release:** certificates, fallback paths, and audit continuity are observed rather than assumed, so runtime safety is measured as a released-action discipline instead of an undocumented filter.

The combined runtime-governance table also makes the evidence boundary easy to read. Battery is the witness row. AV, industrial, and healthcare are defended bounded rows with full certificate rates and bounded runtime semantics. Navigation and aerospace are present in the runtime surface, but they remain blocked from defended closure for data and task reasons rather than architecture reasons. That is exactly the disciplined claim shape a professor-facing draft should preserve.

What this means scientifically is that ORIUS is being evaluated as a release layer, not as a task policy. The battery witness demonstrates the deepest theorem-to-runtime closure, but the runtime-governance table shows the more general point: intervention, fallback, latency, and certificate validity can be measured under one contract across qualitatively different physical systems. That is why the paper uses bounded rows and a witness row rather than forcing every domain into one artificial leaderboard.

The deployment-scope table prevents the results section from sliding into unbounded deployment rhetoric. Even the strongest battery row stops short of unrestricted field deployment. The defended bounded rows stop at replay-backed and runtime-governed validation. The blocked rows remain visible precisely so that the paper's universal architecture claim stays larger than a single benchmark number but smaller than a universal deployment claim.

One further interpretive point matters for the professor review draft. ORIUS is not claiming that every domain should optimize the same surface metric. Some rows care about dispatch legality, others about headway or envelope preservation, others about intervention legality. The universal result is instead that each domain can expose a true-state safety object, and that ORIUS can reduce risk on that object by governing release under degraded observation. That is the empirical thesis the paper can defend without overstating parity.

IX. Cross-Domain Synthesis

ORIUS is best read as a universal runtime safety layer for Physical AI, not as a single-domain controller that happens to generalize. The common object across the monograph is the degraded-observation hazard: the controller sees an observation surface, but safety is judged on the true state or the observation-consistent state set. That is why

TABLE V: Runtime and governance summary for the professor-facing ORIUS draft.

Domain	Row	P95 latency (ms)	Repair (%)	Fallback	Audit (%)	Current limit
Battery Energy Storage	Witness row	0.1	n/a_reference	safe_hold	97.0	external bench and field telemetry packaging still remain bounded deployment items
Autonomous Vehicles	Defended bounded row	0.2	100.0	full_brake	93.0	promoted
Industrial Process Control	Defended bounded row	0.2	91.7	power_cap	94.0	industrial_train_validatio
Medical and Healthcare Monitoring	Defended bounded row	0.1	16.7	max_alert	93.0	healthcare_train_validatio
Navigation and Guidance	Shadow-synthetic	0.1	100.0	hold_position	58.0	navigation_real_data_gap
Aerospace Control	Experimental	0.2	18.8	envelope_hold	55.0	aerospace_experimental_p

the same runtime grammar recurs across battery storage, autonomous vehicles, industrial process control, healthcare monitoring, navigation, and aerospace.

The synthesis that follows from this status is narrower and stronger than a generic transfer-learning claim. ORIUS standardizes the runtime objects that matter for safety under degraded observation: reliability assessment, uncertainty widening, admissible-set tightening, repair or fallback, certificate emission, and auditable lifecycle logging. What varies across domains is the safety geometry, the telemetry parser, and the domain-specific repair policy. What stays invariant is the safety-layer grammar.

That distinction is the key cross-domain result. Battery remains the witness row, but it does not monopolize the theory. Industrial and healthcare are defended bounded rows with real replay evidence. Autonomous vehicles are defended under the current bounded TTC contract. Navigation and aerospace are still informative because they show where the parity gate has not yet been earned.

The professor-facing takeaway is therefore precise: ORIUS already supports a universal safety architecture, but the evidence ladder is still tiered. The architecture is universal in contract; the closure depth is not yet universal in proof. That is exactly the right level of honesty for an R1-grade monograph.

A. Why ORIUS is more than a multi-domain wrapper

Many systems can be described across multiple domains if the description remains qualitative enough. ORIUS aims at a stronger standard. It keeps the same runtime question, the same typed release objects, and the same benchmark semantics across all rows. That is why the framework should be understood as universal in *runtime logic* rather than merely broad in *application count*. The universal claim does not come from listing six domains. It comes from showing that the same degraded- observation semantics survive contact with six different action geometries.

B. Autonomous vehicles

The AV row treats degraded observation as a time-to-collision and entry-barrier hazard under bounded vehicle semantics. Its runtime object is not the full autonomy stack. It is the release decision under uncertain headway and environment state. The defended bounded-row result matters because it shows the ORIUS contract moving into a fast-reacting closed-loop domain without importing battery-specific actuation logic. The row is bounded intentionally: the claim is about governed braking or maneuver legality, not full autonomy closure.

C. Industrial process control

The industrial row treats degraded observation as envelope-preservation risk under process telemetry loss or drift. The safety object is bounded operation inside process constraints with explicit fallback semantics. This row demonstrates that ORIUS can govern a plant whose action geometry and monitoring cadence look nothing like a storage asset. Its universality contribution is that the repair operator can be phrased as power-cap or envelope-preserving modification rather than storage dispatch logic.

D. Healthcare monitoring

The healthcare row is the clearest reminder that ORIUS is not only about actuation- heavy control. Here the danger is intervention suppression or alert illegality under degraded vital-sign observation. The defended bounded-row result shows that the same runtime grammar still applies when the safety object is intervention legality rather than dispatch power. It also marks a clean scientific boundary: the paper is not claiming bedside autonomy certification, only monitored intervention legality under a bounded runtime contract.

E. Navigation

The navigation row frames degraded observation as corridor or trajectory-feasibility risk under degraded localization. The adapter and replay harness already exist, which means the runtime contract is portable. What is missing is the defended real-data closure row. That is why navigation remains shadow-synthetic in the present paper.

F. Aerospace

The aerospace row frames degraded observation as flight-envelope preservation under degraded sensing and prognostic uncertainty. Its current role is boundary-setting:

TABLE VI: Deployment validation scope for ORIUS. The table distinguishes defended replay, rehearsal, proxy, and explicitly out-of-scope surfaces so deployment language cannot outrun the tracked artifacts.

Deployment surface	Governing artifact	Scope type	Current status	Bounded manuscript claim	Exact non-claim or gap
Battery runtime and HIL rehearsal	reports/publication/dcshearcyclesummary.csv; reports/hil/hil_summary.json	rehearsal	bounded_reference	Battery supports the deepest runtime and HIL-like evidence in the book.	This is still not a full external bench or field deployment package.
Battery aging and half-life	reports/aging/asset_proxy_plus_design_table.csv; reports/publication/aging_aware_calibration_design.md	proxy	partial	The book can defend proxy-backed aging and half-life design reasoning.	It does not yet defend a full live aging-validation stack.
Autonomous-vehicle defended replay	reports/universal_orius_validation/domain_defended_matrix.csv	bounded replay	bounded	AV can claim bounded proof-validated replay under the TTC entry-barrier contract.	It cannot claim full live-stack road deployment or universal AV closure.
Industrial and healthcare defended replay	reports/universal_orius_validation/domain_defended_matrix.csv	bounded replay	bounded	Industrial and healthcare can claim bounded defended replay plus runtime governance.	They do not claim universal plant-family closure or regulated clinical deployment.
Navigation field telemetry	data/navigation/PLACE_REAL_NAVIGATION_DATA_HERE.md; reports/publication/orius_equal_domain_parity_matrix.csv	real-data requirement	blocked	Navigation remains a portability row only.	Real KITTI-backed train, replay, soundness, and runtime traces are still missing.
Aerospace flight telemetry	data/aerospace/PLACE_REAL_AEROSPACE_DATA_HERE.md; reports/publication/orius_equal_domain_parity_matrix.csv	real-flight requirement	blocked	Aerospace remains an experimental outer-boundary row.	A defended real multi-flight task and replay improvement are still missing. A bounded public ADS-B runtime support lane exists but cannot promote the official row.
OOD and adversarial completeness	chapters/ch34_outside_explicit_non_claim.tex; reports/publication/adversarial_probing_robustness_table.csv	explicit non-claim	registered	The monograph can discuss bounded active probing and non-claim discipline.	It does not claim universal adversarial completeness or unrestricted OOD safety.

the runtime objects and fallback semantics are already expressible, but the defended multi-flight validation surface is not yet closed. Keeping aerospace visible but gated is scientifically stronger than overstating parity.

X. Explicit Evidence Boundary

The manuscript does not overrun repository truth. The current repo supports a universal runtime-safety architecture, but it does not support a flat claim of equal empirical maturity across all six domains. The evidence boundary is therefore part of the scientific contribution, not a defensive footnote.

The tracked matrices make the boundary explicit. In `reports/publication/orius_equal_domain_parity_matrix.csv`, battery is the reference witness, autonomous vehicles, industrial process control, and healthcare monitoring are defended bounded rows, navigation is shadow-synthetic, and aerospace is experimental. The runtime-budget matrix and the universal validation summary say the same thing in a different form: AV, industrial, and healthcare clear their bounded contracts; navigation remains blocked by the missing real-data row; and aerospace remains gated by the lack of a real multi-flight safety task with material post-repair improvement.

The paper therefore uses the following claim discipline:

- It may claim a universal safety layer for Physical AI under degraded observation.
- It may claim bounded validation for AV, industrial, and healthcare under their current contracts.
- It may claim portability and protocol design for navigation.
- It may claim experimental boundary evidence for aerospace.

It does not claim equal-domain universality, universal hardware closure, or field deployment readiness for the navigation and aerospace rows. Those would outrun the present repository evidence and the locked validation outputs.

TABLE VII: Cross-domain parity and promotion status for the professor-facing ORIUS draft.

Domain	Current row	What is currently closed	Current blocker or boundary
Battery Energy Storage	Witness row	adapter, replay, CertOS, soundness	battery_reference_witness
Autonomous Vehicles	Defended bounded row	adapter, replay, CertOS, soundness	promoted
Industrial Process Control	Defended bounded row	adapter, replay, CertOS, soundness	industrial_train_validation_chain_complete
Medical and Healthcare Monitoring	Defended bounded row	adapter, replay, CertOS, soundness	healthcare_train_validation_chain_complete
Navigation and Guidance	Shadow-synthetic	adapter, replay	navigation_real_data_gap
Aerospace Control	Experimental	adapter, replay	aerospace_experimental_placeholder

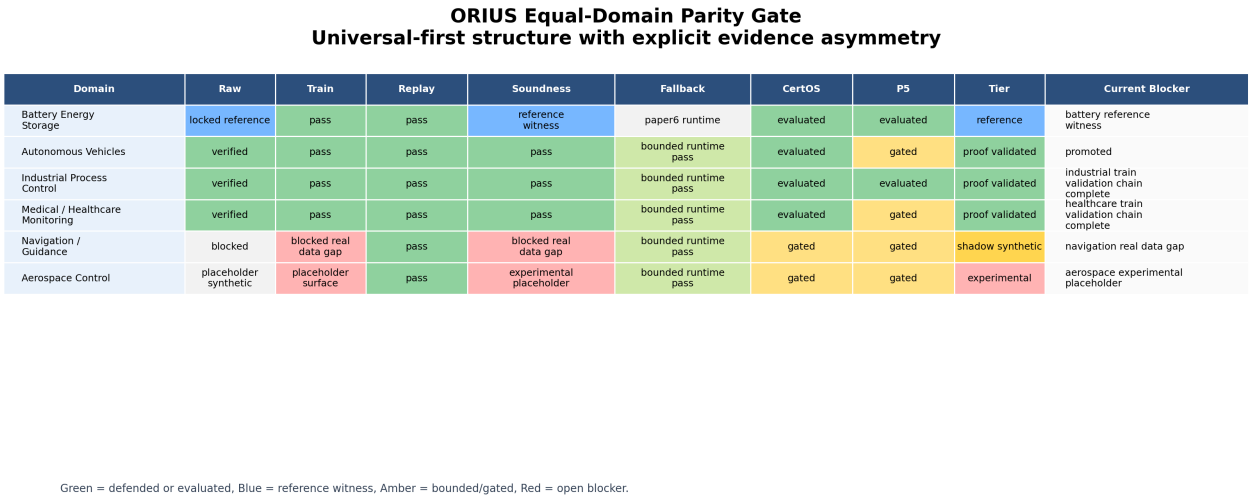

Fig. 5: Parity-gated view of the ORIUS program. The architecture is universal now, but domain promotion remains governed by closure status rather than by rhetorical symmetry.

TABLE VIII: Allowed claims versus non-claims at the current evidence boundary.

Claim family	Supported by current repo truth	Not supported yet
Universal runtime grammar	One adapter contract, one benchmark schema, one CertOS lifecycle, and one degraded-observation hazard model across all rows.	A claim that all rows have identical proof depth or equal closure.
Battery witness depth	Deep theorem-to-code-to-artifact lineage with locked reference artifacts.	Treating battery as the conceptual center of the paper.
AV / industrial / healthcare	Defended bounded rows with locked replay and bounded runtime semantics.	Unqualified claims of unrestricted deployment readiness.
Navigation	Portability and protocol design evidence with a visible real-data gap.	Field-grade navigation validation or parity with the defended rows.
Aerospace	Experimental boundary evidence plus supplemental runtime diagnostics.	Proof-candidate or proof-validated aerospace closure.

The practical implication is straightforward. The manuscript can be ambitious about architecture, but it must stay conservative about closure. That is not a weakness in the ORIUS program. It is the reason the program remains defensible under serious review.

XI. Conclusion

ORIUS argues that the right universal object for Physical AI safety is not a single controller, a single benchmark, or a single proof family. It is a runtime safety layer that mediates degraded observation before actuation. Once telemetry reliability becomes part of the safety problem, the system must expose reliability, uncertainty inflation, tightened action legality, repair, fallback, and certificate semantics as first-class objects. That is the contribution of the Detect–Calibrate–Constrain–Shield–Certify kernel.

The manuscript makes that contribution at three levels. First, it defines OASG as the dominant hidden hazard: actions can remain legal on the observed state while becoming unsafe on the true state. Second, it shows how a universal runtime contract can bind adapter semantics, benchmark semantics, and CertOS governance under one typed layer. Third, it demonstrates that the architecture is more than a diagram: the battery row provides witness-

depth theorem-to-code-to-artifact closure, and the defended bounded rows show that the same runtime grammar travels beyond energy.

The scientific boundary is intentionally explicit. The paper does not claim equal-domain closure. Navigation remains blocked by missing defended real-data replay, and aerospace remains blocked by the absence of a real multi-flight runtime validation surface with material post-repair gain. Those blockers do not weaken the architectural thesis. They show that ORIUS is disciplined enough to separate universality of contract from equality of evidence.

What changes at the field level is the object of research. ORIUS suggests that Physical AI safety should be organized around reusable runtime layers that sit between nominal intelligence and physical release. Such a layer can host conformal uncertainty, safety filters, runtime assurance, and anomaly detection without collapsing into any one of them. If the remaining blocked rows are closed, ORIUS supports a stronger universal claim. Even before that point, it establishes a credible safety-layer grammar for Physical AI under degraded observation.

In that sense, ORIUS is also a writing and evaluation proposal for the field. It argues that future physical-AI systems should report not only predictive accuracy or planner reward, but also true-state violation behavior, intervention frequency, fallback usage, certificate validity, and release latency under degraded observation. That reporting discipline matters because many failures in real systems are not failures of nominal optimization; they are failures of release semantics once observation ceases to be trustworthy. A framework like ORIUS makes those failures measurable and governable.

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